Learned Beltrami–Laplacian Neural Operators on Compact Manifolds

Fixed-Metric Approximation, Manifold Aliasing, and Spectral Optimality

Abstract

Neural operators learn maps between function spaces. Fourier neural operators use global Fourier modes and are therefore naturally adapted to periodic Euclidean domains or flat tori. This document develops a fixed-metric Beltrami–Laplacian analogue on compact Riemannian manifolds. The Fourier basis is replaced by the eigenbasis of the Laplace–Beltrami operator. We prove compact-set universal approximation for continuous operators, a quantitative truncation-plus-network error decomposition under explicit Lipschitz and moment assumptions, a manifold pseudo-spectral aliasing bound, and the optimality of Laplace–Beltrami eigenfunctions for linear approximation of Sobolev balls in L^2.

The results are deliberately stated for fixed realised metrics. Learned metrics are treated as an optimisation mechanism for finding task-adapted spectral coordinates rather than as a claim of unique recovery of an underlying geometry. This distinction is important: inverse spectral geometry shows that a metric need not be uniquely determined by spectral information. The relevant approximation claim is therefore not that training discovers the unique correct metric, but that it may discover a dynamically adequate metric whose eigenbasis compactly represents the operator of interest.

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1 Introduction and setup

1.1 Motivation

Neural operators learn mappings between infinite-dimensional function spaces. The Fourier Neural Operator (FNO) [15] works by projecting onto truncated Fourier modes, applying a learned finite-dimensional map in coefficient space, and reconstructing. This is natural on periodic Euclidean domains because Fourier modes diagonalise the Laplacian on the flat torus.

On a general compact Riemannian manifold there is no global Fourier basis. The Laplace–Beltrami eigenfunctions provide the intrinsic replacement: they form a complete orthonormal basis of L^2, diagonalise the Laplace–Beltrami operator, and obey Weyl eigenvalue asymptotics. The present construction replaces Fourier modes by Beltrami–Laplacian modes.

The basic architecture is

$$\widehat{\mathcal{G}} = R^Y_{N_Y} \circ G_{\mathrm{NN}} \circ P^X_{N_X}, \tag{1}$$

where $P^X_{N_X}$ projects the input onto the first N_X Laplace–Beltrami modes on X, G_{NN} is a finite-dimensional neural network in spectral coefficient space, and $R^Y_{N_Y}$ reconstructs from the first N_Y modes on Y.

The central purpose of this document is to state precisely what this construction proves without relying on the stronger Fourier-specific machinery used in toroidal settings. The theorem below is less restrictive topologically than requiring a torus: it applies to compact closed Riemannian manifolds. In practical learned-geometry settings, however, a single global coordinate representation may not exist or may be numerically poor. More general spaces may require an atlas of local coordinate charts, with learned metric tensors on each chart and compatibility conditions on chart overlaps. This is a modelling and implementation issue rather than a restriction of the fixed-metric spectral theorem.

1.2 Manifolds and the Laplace–Beltrami operator

Throughout, (X^{n_X}, g_X) and (Y^{n_Y}, g_Y) are compact, connected, smooth Riemannian manifolds without boundary. The dimensions n_X and n_Y may differ.

For a compact closed Riemannian manifold (M^n, g), the Laplace–Beltrami operator is

$$\Delta_g u = |g|^{-1/2} \partial_i \left(|g|^{1/2} g^{ij} \partial_j u \right). \tag{2}$$

We use the convention that $-\Delta_g$ is non-negative. Its eigenvalues are written

$$0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \cdots \to +\infty, \tag{3}$$

with corresponding real L^2-orthonormal eigenfunctions $\{\phi_k\}_{k=0}^{\infty}$.

Remark 1.1 (Manifolds with boundary)**.** *The main results are stated for closed manifolds to avoid boundary-condition notation. Analogues hold for compact manifolds with smooth boundary once Dirichlet or Neumann boundary conditions are fixed. Without such a choice, the Laplace–Beltrami operator admits multiple self-adjoint realisations. Boundary conditions therefore form part of the operator definition.*

1.3 Mode-count convention

We use N to mean the number of retained modes. Thus

$$P_N u = \sum_{k=0}^{N-1} \langle u, \phi_k \rangle_{L^2(M)} \phi_k. \tag{4}$$

This convention includes the constant mode ϕ_0. If one works on zero-mean spaces, the sum may instead begin at $k = 1$, but that is a different convention and must be stated separately.

The coefficient encoder and synthesis decoder are

$$P_N^M u = (\langle u, \phi_0 \rangle, \dots, \langle u, \phi_{N-1} \rangle) \in \mathbb{R}^N, \tag{5}$$

$$R_N^M a = \sum_{k=0}^{N-1} a_k \phi_k. \tag{6}$$

The synthesis map $R_N^M : \ell_N^2 \to L^2(M)$ is an isometry. It is not an isometry into $H^s(M)$ unless the coefficient norm is weighted by $(1 + \lambda_k)^{s/2}$.

1.4 Sobolev spaces

For $s \geq 0$, define the spectral Sobolev norm

$$\|u\|_{H^s(M)}^2 = \sum_{k=0}^{\infty} (1 + \lambda_k)^s |a_k|^2, \qquad a_k = \langle u, \phi_k \rangle_{L^2(M)}. \tag{7}$$

On compact smooth manifolds this norm is equivalent to the usual chart-based Sobolev norm, with constants depending on the geometry [7, 8, 4]. For typographical consistency in the bibliography we cite Hebey as [7].

1.5 Target operators

The basic target is a continuous operator

$$\mathcal{G} : H^s(X) \to H^t(Y). \tag{8}$$

For the compact-set existence result, continuity in the H^s topology is sufficient. For the quantitative truncation bound, we impose an explicit L^2-Lipschitz condition on the relevant input class:

$$\|\mathcal{G}(u) - \mathcal{G}(v)\|_{L^2(Y)} \leq L_{\mathcal{G}} \|u - v\|_{L^2(X)}. \tag{9}$$

This distinction is essential. Continuity in H^s does not imply continuity with respect to the weaker L^2 metric.

2 Spectral foundations

Lemma 2.1 (Self-adjointness and spectral decomposition)**.** *Let (M^n, g) be a compact closed Riemannian manifold. Then $-\Delta_g$ admits a unique self-adjoint realisation on $L^2(M)$ with domain $H^2(M)$. Its spectrum is discrete,*

$$0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \cdots \to +\infty, \tag{10}$$

with finite multiplicities, and the eigenfunctions form a complete orthonormal basis of $L^2(M)$.

Proof. Define the quadratic form

$$\mathfrak{a}(u, v) = \int_M g(\nabla_g u, \nabla_g v) \, \mathrm{d}V_g, \qquad \mathrm{dom}(\mathfrak{a}) = H^1(M). \tag{11}$$

It is densely defined, symmetric, non-negative, and closed with respect to the form norm

$$\|u\|_{\mathfrak{a}}^2 = \|u\|_{L^2}^2 + \mathfrak{a}(u, u), \tag{12}$$

which is equivalent to the usual H^1 norm. By the representation theorem for closed non-negative symmetric forms, there is a unique non-negative self-adjoint operator A satisfying

$$\langle Au, v\rangle_{L^2} = \mathfrak{a}(u, v) \tag{13}$$

for all $u \in \mathrm{dom}(A)$ and $v \in H^1(M)$ [9]. This operator is the Friedrichs realisation of $-\Delta_g$.

On a smooth closed manifold, elliptic regularity gives $\mathrm{dom}(A) = H^2(M)$ and $A = -\Delta_g$ in the distributional sense [4]. The resolvent $(A + I)^{-1} : L^2(M) \to H^2(M)$ is bounded, while the embedding $H^2(M) \hookrightarrow L^2(M)$ is compact by Rellich–Kondrachov. Hence $(A + I)^{-1}$ is a compact self-adjoint operator on $L^2(M)$. The spectral theorem for compact self-adjoint operators gives a complete orthonormal eigenbasis with eigenvalues accumulating only at zero for the resolvent, equivalently eigenvalues tending to $+\infty$ for A [10].

The constant functions lie in the kernel. Since M is connected, the maximum principle implies that every harmonic function is constant, so the zero eigenvalue has multiplicity one and $\lambda_1 > 0$. □

Lemma 2.2 (Spectral truncation)**.** *Let $u \in H^s(M)$, $s > 0$. Then*

$$\|u - P_N u\|^2_{L^2(M)} = \sum_{k=N}^{\infty} |a_k|^2 \le (1 + \lambda_N)^{-s} \|u\|^2_{H^s(M)}, \tag{14}$$

where P_N retains the modes $k = 0, \dots, N - 1$.

Proof. By Parseval,

$$\|u - P_N u\|^2_{L^2} = \sum_{k=N}^{\infty} |a_k|^2. \tag{15}$$

For $k \ge N$, $\lambda_k \ge \lambda_N$, and therefore

$$\sum_{k=N}^{\infty} |a_k|^2 = \sum_{k=N}^{\infty} \frac{(1 + \lambda_k)^s |a_k|^2}{(1 + \lambda_k)^s} \le (1 + \lambda_N)^{-s} \sum_{k=N}^{\infty} (1 + \lambda_k)^s |a_k|^2. \tag{16}$$

This is bounded by $(1 + \lambda_N)^{-s} \|u\|^2_{H^s}$. □

Lemma 2.3 (Weyl asymptotics)**.** *On a compact closed Riemannian manifold (M^n, g),*

$$\lambda_k \sim (2\pi)^2 \left(\omega_n \,\mathrm{vol}(M, g)\right)^{-2/n} k^{2/n} \qquad \text{as } k \to \infty, \tag{17}$$

where ω_n is the volume of the Euclidean unit ball in $\mathbb{R}^n$.

Proof. This is Weyl's law for the Laplace–Beltrami operator; see Weyl [5] and Hörmander [6]. □

Corollary 2.4 (Sobolev–Weyl truncation rate)**.** *Let $u \in H^s(M^n)$ with $\|u\|_{H^s} \le R$. Then for all sufficiently large N,*

$$\|u - P_N u\|_{L^2(M)} \le C(M, g, s) R N^{-s/n}. \tag{18}$$

Proof. By (14),

$$\|u - P_N u\|^2_{L^2} \le (1 + \lambda_N)^{-s} R^2. \tag{19}$$

Weyl's law implies that for N sufficiently large, $\lambda_N \ge cN^{2/n}$. Thus

$$\|u - P_N u\|_{L^2} \le C R N^{-s/n}. \tag{20}$$

□

Lemma 2.5 (Sobolev embedding and projection contraction)**.** *If $s > n/2$, then*

$$H^s(M) \hookrightarrow L^\infty(M), \qquad \|u\|_{L^\infty} \le C_S(M, g, s)\|u\|_{H^s}. \tag{21}$$

Moreover,

$$\|P_N u\|_{H^s} \le \|u\|_{H^s}. \tag{22}$$

Proof. The embedding is the standard Sobolev embedding theorem on compact manifolds [7, 8]. The contraction follows directly from the spectral definition:

$$\|P_N u\|_{H^s}^2 = \sum_{k=0}^{N-1} (1+\lambda_k)^s |a_k|^2 \le \sum_{k=0}^{\infty} (1+\lambda_k)^s |a_k|^2. \tag{23}$$

□

3 Fixed-metric universal approximation

Lemma 3.1 (Finite-dimensional universal approximation)**.** *Let $K \subset \mathbb{R}^d$ be compact and let $f : K \to \mathbb{R}^m$ be continuous. For every $\varepsilon > 0$, there exists a ReLU network $\psi : \mathbb{R}^d \to \mathbb{R}^m$ such that*

$$\sup_{x \in K} |f(x) - \psi(x)| \le \varepsilon. \tag{24}$$

Proof. This is the standard universal approximation theorem for non-polynomial activation functions, specialised to ReLU networks; see Cybenko [11], Hornik–Stinchcombe–White [12], Leshno et al. [13], and Pinkus [14]. □

Theorem 3.2 (Compact-set existence)**.** *Let (X, g_X) and (Y, g_Y) be compact closed Riemannian manifolds. Let $s, t \ge 0$, and let*

$$\mathcal{G} : H^s(X) \to H^t(Y) \tag{25}$$

be continuous. For every compact $K \subset H^s(X)$ and every $\varepsilon > 0$, there exist truncation levels N_X, N_Y and a ReLU network

$$G_{\mathrm{NN}} : \mathbb{R}^{N_X} \to \mathbb{R}^{N_Y} \tag{26}$$

for which

$$\sup_{u \in K} \left\| \mathcal{G}(u) - R^Y_{N_Y} G_{\mathrm{NN}} P^X_{N_X} u \right\|_{L^2(Y)} \le \varepsilon. \tag{27}$$

Proof. Fix $\varepsilon > 0$ and compact $K \subset H^s(X)$.

First, $\mathcal{G}(K)$ is compact in $H^t(Y)$ and therefore compact in $L^2(Y)$. Since $P^Y_N \to \mathrm{Id}$ strongly in $L^2(Y)$ and the projections are uniformly bounded, the convergence is uniform on compact subsets. Hence there exists N_Y such that

$$\sup_{v \in \mathcal{G}(K)} \|v - P^Y_{N_Y} v\|_{L^2(Y)} \le \varepsilon/3. \tag{28}$$

Second, $P^X_N \to \mathrm{Id}$ strongly in $H^s(X)$ and the projections are contractions in $H^s(X)$. Strong convergence of a uniformly bounded sequence of linear operators is uniform on compact sets. Thus

$$\sup_{u \in K} \|u - P^X_{N_X} u\|_{H^s(X)} \tag{29}$$

can be made arbitrarily small. Since $\mathcal{G}$ is continuous on the compact set K and $P^X_{N_X} K$ lies in an arbitrarily small H^s-neighbourhood of K for large N_X, uniform continuity on a compact neighbourhood gives

$$\sup_{u \in K} \|\mathcal{G}(u) - \mathcal{G}(P^X_{N_X} u)\|_{L^2(Y)} \le \varepsilon/3 \tag{30}$$

for sufficiently large N_X.

Define the finite-dimensional latent map

$$\chi(a) = P^Y_{N_Y}\mathcal{G}(R^X_{N_X}a), \qquad a \in P^X_{N_X}(K) \subset \mathbb{R}^{N_X}. \tag{31}$$

This map is continuous on the compact set $P^X_{N_X}(K)$. By finite-dimensional universal approximation, choose G_{NN} such that

$$\sup_{a\in P^X_{N_X}(K)} |\chi(a) - G_{\mathrm{NN}}(a)| \le \varepsilon/3. \tag{32}$$

Since $R^Y_{N_Y} : \ell^2_{N_Y} \to L^2(Y)$ is an isometry, this gives

$$\sup_{u\in K} \|R^Y_{N_Y}\chi(P^X_{N_X}u) - R^Y_{N_Y}G_{\mathrm{NN}}P^X_{N_X}u\|_{L^2(Y)} \le \varepsilon/3. \tag{33}$$

Combining the output truncation, input truncation, and network approximation errors proves (27). □

Remark 3.3 (Why the topology matters)**.** *The proof uses compactness and continuity in the H^s topology. It does not assert that H^s-continuity implies L^2-continuity. That implication is false in general because L^2 is a weaker topology than H^s.*

4 Quantitative fixed-metric error bounds

Lemma 4.1 (Three-way decomposition)**.** *Let $\mathcal{G}$ satisfy the L^2-Lipschitz condition (9). Let*

$$\widehat{\mathcal{G}} = R^Y_{N_Y} \circ G_{\mathrm{NN}} \circ P^X_{N_X}. \tag{34}$$

Define

$$\chi(a) = P^Y_{N_Y}\mathcal{G}(R^X_{N_X}a). \tag{35}$$

Then, for any probability measure μ on $H^s(X)$,

$$\begin{aligned}\mathbb{E}_\mu\|\mathcal{G}(u) - \widehat{\mathcal{G}}(u)\|^2_{L^2(Y)} \le 3\Big(&L^2_{\mathcal{G}}\mathbb{E}_\mu\|u - P^X_{N_X}u\|^2_{L^2(X)} \\ &+ \mathbb{E}_\mu|G_{\mathrm{NN}}(P^X_{N_X}u) - \chi(P^X_{N_X}u)|^2 \\ &+ \mathbb{E}_\mu\|\mathcal{G}(u) - P^Y_{N_Y}\mathcal{G}(u)\|^2_{L^2(Y)}\Big).\end{aligned} \tag{36}$$

Proof. For each u, write

$$\begin{aligned}\widehat{\mathcal{G}}(u) - \mathcal{G}(u) = &\left[R^Y_{N_Y}G_{\mathrm{NN}}P^X_{N_X}u - R^Y_{N_Y}\chi(P^X_{N_X}u)\right] \\ &+ \left[R^Y_{N_Y}\chi(P^X_{N_X}u) - P^Y_{N_Y}\mathcal{G}(u)\right] \\ &+ \left[P^Y_{N_Y}\mathcal{G}(u) - \mathcal{G}(u)\right].\end{aligned} \tag{37}$$

The first term is controlled by the coefficient error because $R^Y_{N_Y}$ is an L^2 isometry on $\ell^2_{N_Y}$. The second term is

$$P^Y_{N_Y}\mathcal{G}(P^X_{N_X}u) - P^Y_{N_Y}\mathcal{G}(u), \tag{38}$$

which is bounded in $L^2(Y)$ by

$$\|\mathcal{G}(P^X_{N_X}u) - \mathcal{G}(u)\|_{L^2(Y)} \le L_{\mathcal{G}}\|P^X_{N_X}u - u\|_{L^2(X)}. \tag{39}$$

The third term is the output spectral tail. Finally use

$$\|a + b + c\|^2 \le 3(\|a\|^2 + \|b\|^2 + \|c\|^2) \tag{40}$$

and integrate with respect to μ. □

Theorem 4.2 (Quantitative fixed-metric bound)**.** *Let (X^{n_X}, g_X) and (Y^{n_Y}, g_Y) be compact closed Riemannian manifolds. Let*

$$\mathcal{G} : H^s(X) \to H^t(Y) \tag{41}$$

be measurable and satisfy the L^2-Lipschitz condition (9) on the relevant input class. Let μ be a probability measure on $H^s(X)$ such that

$$\mathbb{E}_\mu \|u\|^2_{H^s(X)} \le B, \qquad E^Y_\mu := \mathbb{E}_\mu \|\mathcal{G}(u)\|^2_{H^t(Y)} < \infty. \tag{42}$$

Then

$$\mathbb{E}_\mu \|\mathcal{G}(u) - \widehat{\mathcal{G}}(u)\|^2_{L^2(Y)} \le 3\Big(L^2_{\mathcal{G}} C_X^2 B N_X^{-2s/n_X} + \varepsilon^2_{\mathrm{NN}} + C_Y^2 E^Y_\mu N_Y^{-2t/n_Y} \Big), \tag{43}$$

where

$$\varepsilon^2_{\mathrm{NN}} := \mathbb{E}_\mu |G_{\mathrm{NN}}(P^X_{N_X} u) - P^Y_{N_Y} \mathcal{G}(P^X_{N_X} u)|^2. \tag{44}$$

The constants C_X and C_Y are the Sobolev–Weyl constants of X and Y.

Proof. Apply the three-way decomposition (36). By the Sobolev–Weyl truncation estimate,

$$\mathbb{E}_\mu \|u - P^X_{N_X} u\|^2_{L^2(X)} \le C_X^2 N_X^{-2s/n_X} \mathbb{E}_\mu \|u\|^2_{H^s(X)} \le C_X^2 B N_X^{-2s/n_X}. \tag{45}$$

Similarly,

$$\mathbb{E}_\mu \|\mathcal{G}(u) - P^Y_{N_Y} \mathcal{G}(u)\|^2_{L^2(Y)} \le C_Y^2 N_Y^{-2t/n_Y} \mathbb{E}_\mu \|\mathcal{G}(u)\|^2_{H^t(Y)}. \tag{46}$$

Substitution gives (43). □

Remark 4.3 (Network approximation term)**.** *The quantity $\varepsilon_{\mathrm{NN}}$ is the finite-dimensional neural-network approximation error for the latent map. It may be bounded using finite-dimensional approximation theory if additional smoothness of the latent map is known. For a merely Lipschitz latent map, no dimension-free parameter-count rate should be inferred without further structure.*

5 Manifold pseudo-spectral aliasing

The following estimate is the intrinsic analogue of a basic Fourier pseudo-spectral aliasing bound. It does not require a global Fourier basis and does not estimate individual triple products of eigenfunctions.

Lemma 5.1 (Bilinear manifold aliasing)**.** *Let (M^n, g) be compact closed and let $s > n/2$. For $f, h \in H^s(M)$,*

$$\|P_N(fh) - P_N(P_N f\, P_N h)\|_{L^2(M)} \le 2 C_W C_S N^{-s/n} \|f\|_{H^s} \|h\|_{H^s}, \tag{47}$$

for all sufficiently large N, where C_W is the Sobolev–Weyl truncation constant and C_S is the Sobolev embedding constant.

Proof. Since P_N is an orthogonal projection on L^2,

$$\|P_N(fh) - P_N(P_N f\, P_N h)\|_{L^2} \le \|fh - P_N f\, P_N h\|_{L^2}. \tag{48}$$

Using

$$fh - P_N f\, P_N h = (f - P_N f)h + P_N f(h - P_N h), \tag{49}$$

we obtain

$$\|fh - P_N f\, P_N h\|_{L^2} \le \|f - P_N f\|_{L^2}\|h\|_{L^\infty} + \|P_N f\|_{L^\infty}\|h - P_N h\|_{L^2}. \tag{50}$$

By Sobolev embedding and projection contraction,

$$\|h\|_{L^\infty} \le C_S\|h\|_{H^s}, \qquad \|P_N f\|_{L^\infty} \le C_S\|P_N f\|_{H^s} \le C_S\|f\|_{H^s}. \tag{51}$$

By Sobolev–Weyl truncation,

$$\|f - P_N f\|_{L^2} \le C_W N^{-s/n}\|f\|_{H^s}, \qquad \|h - P_N h\|_{L^2} \le C_W N^{-s/n}\|h\|_{H^s}. \tag{52}$$

Combining these estimates gives (47). □

Corollary 5.2 (Polynomial products)**.** *Let $s > n/2$ and $f_1, \dots, f_p \in H^s(M)$. Then*

$$\left\| P_N\left(\prod_{i=1}^p f_i\right) - P_N\left(\prod_{i=1}^p P_N f_i\right)\right\|_{L^2} \le pC_W C_S^{p-1} N^{-s/n} \prod_{i=1}^p \|f_i\|_{H^s}. \tag{53}$$

Proof. Use the telescoping identity

$$\prod_{i=1}^p f_i - \prod_{i=1}^p P_N f_i = \sum_{j=1}^p \left(\prod_{i<j} P_N f_i\right)(f_j - P_N f_j)\left(\prod_{i>j} f_i\right). \tag{54}$$

Each term has one L^2 tail factor and $p-1$ L^∞ factors. Apply Sobolev embedding, projection contraction, and the truncation estimate. □

Remark 5.3 (Why individual eigenfunction bounds are not needed)**.** *One could try to estimate aliasing using coefficients of the form $\int_M \phi_i\phi_j\phi_k \, \mathrm{d}V_g$. This would introduce individual eigenfunction estimates such as Sogge's L^p bounds [17]. The proof above avoids this route. It controls the whole product in Sobolev and L^∞ norms, so the rate is governed by Sobolev regularity and Weyl asymptotics rather than by pointwise growth of individual eigenfunctions.*

6 Structured PDE example: Darcy flow

The aliasing lemma explains why the Beltrami–Laplacian basis is a natural manifold analogue of Fourier pseudo-spectral methods. Consider, formally, an elliptic equation

$$-\nabla_g \cdot (a\nabla_g u) = f, \qquad a = 1 + \tilde{a}, \tag{55}$$

with suitable compatibility conditions on a closed manifold. In the flat periodic case, Fourier pseudo-spectral methods approximate products such as $\tilde{a}\nabla u$ by transforming between coefficient and physical representations. Lemma 47 gives the scalar manifold analogue for products in H^s.

A full Darcy emulation theorem on a manifold would require more than the scalar aliasing bound. One would need vector-valued aliasing estimates for terms such as $\tilde{a}\nabla_g v$, mapping estimates for operators such as $(-\Delta_g)^{-1}\nabla_g\cdot$, and elliptic regularity assumptions strong enough to control the coefficient field and solution in the required Sobolev spaces. These are operator-specific analytic requirements. They are not needed for the fixed-metric approximation theorem in this document.

Thus Darcy flow is best viewed here as a structured motivating example: the BL basis supplies the intrinsic spectral coordinates, and the aliasing lemma supplies one of the ingredients needed for pseudo-spectral PDE emulation. The present theorem does not rely on a Darcy-specific rate.

7 Metric learning and non-uniqueness

The results above are fixed-metric approximation results. They apply once a Riemannian metric has been chosen or learned. In applications, one may parameterise a metric g_θ and train it jointly with the finite-dimensional latent map. The mathematical claim in this document is not that training uniquely recovers the true metric. Rather, it is that any realised metric defines a Laplace–Beltrami eigenbasis, and the preceding fixed-metric approximation and truncation results apply to that realised basis.

This distinction is important. Correct metric discovery is an inverse problem, and uniqueness should not be assumed. Inverse spectral geometry shows that spectral data do not in general determine a unique geometry; non-isometric isospectral examples are known [18, 19]. Consequently, a learned metric should be interpreted as task-adapted, dynamically adequate, or operator-adapted unless an additional identifiability theorem is proved.

For the purposes of operator learning, uniqueness is not necessary. If a learned metric produces a spectral basis in which the observed dynamics admit compact, stable, and accurate representation, then it is adequate for the approximation task even if other metrics would produce equivalent performance. This is analogous to metric-learning viewpoints in which the learned geometry encodes useful intrinsic structure rather than certifying unique recovery of a physical manifold.

Remark 7.1 (Perturbation theory)**.** *One may ask how approximation changes when g is perturbed. Such results should be proved using spectral perturbation theory for compact resolvents or quadratic forms. A direct Davis–Kahan argument applied to $\Delta_{\tilde{g}} - \Delta_g : H^2 \to L^2$ is not sufficient, because this is not a bounded self-adjoint perturbation on L^2. Davis–Kahan remains appropriate for bounded perturbations of invariant subspaces, but metric perturbations of elliptic operators require a more careful resolvent or form-based treatment [20, 9, 21].*

Remark 7.2 (Atlases of learned metrics)**.** *The fixed-metric theorem is intrinsic and does not require a global toroidal coordinate system. However, if a metric is learned from local data, some spaces may require an atlas of charts. In such cases one would learn local metric tensors and impose compatibility on overlaps. This is a natural geometric extension, not a change to the spectral theorem itself. The theorem assumes the resulting smooth compact Riemannian manifold and its Laplace–Beltrami operator are already defined.*

8 Spectral optimality and lower bounds

The correct lower-bound statement concerns compression of Sobolev balls, not arbitrary target operators. An arbitrary operator may be zero, finite-rank, or depend only on a small number of input modes, in which case no universal mode-count lower bound is possible.

Definition 8.1 (Kolmogorov width)**.** *For a compact set A in a normed space Z, the Kolmogorov N-width is*

$$d_N(A, Z) = \inf_{\dim V = N} \sup_{a \in A} \inf_{v \in V} \|a - v\|_Z. \tag{56}$$

Theorem 8.2 (Sobolev-ball width)**.** *Let M^n be compact closed and let*

$$B_{H^s(M)} = \{u \in H^s(M) : \|u\|_{H^s} \le 1\}. \tag{57}$$

Then

$$d_N(B_{H^s(M)}, L^2(M)) \asymp N^{-s/n}. \tag{58}$$

Moreover, the span of the first N Laplace–Beltrami eigenfunctions achieves this rate.

Proof. This is the standard Kolmogorov-width estimate for Sobolev balls on compact manifolds, with the upper bound realised by spectral truncation and the lower bound following from the asymptotics of the embedding $H^s(M) \hookrightarrow L^2(M)$; see Pinkus [22] and Pesenson [23]. □

Remark 8.3 (What this lower bound does and does not say)**.** *The theorem proves that the BL eigenbasis is linearly optimal for approximating Sobolev-regular functions in L^2. It does not say that every operator $\mathcal{G} : H^s \to H^t$ requires $N \asymp \varepsilon^{-n/s}$ input modes. Such a statement is false for finite-rank or zero operators. The lower bound should therefore be used to justify the encoder–decoder basis, not as a universal lower bound for all target operators.*

9 Discretisation

In computation, the eigenfunctions are approximated numerically, for example by finite elements or graph-based discretisations. Classical Babuška–Osborn theory gives convergence of eigenvalues and eigenspaces for finite-element approximations of elliptic eigenproblems [24]. For a rigorous discretised version of the present theorem, the projection error must be augmented by a numerical spectral-projector error. In the presence of eigenvalue multiplicities, this should be stated for eigenspaces or spectral projectors rather than individual eigenfunctions.

A typical discretised estimate has the form

$$\|u - P_{N,h}u\|_{L^2} \le \|u - P_N u\|_{L^2} + \|(P_N - P_{N,h})u\|_{L^2}, \tag{59}$$

where $P_{N,h}$ is the numerical spectral projector. The second term depends on mesh quality, polynomial degree, elliptic regularity, and spectral separation. These details are implementation-specific and are not part of the fixed continuous theorem.

10 Main theorem

Theorem 10.1 (Fixed-metric Beltrami–Laplacian neural operator approximation)**.** *Let (X^{n_X}, g_X) and (Y^{n_Y}, g_Y) be compact closed Riemannian manifolds.*

(I) ***Compact-set existence.*** *Let $s, t \ge 0$ and let $\mathcal{G} : H^s(X) \to H^t(Y)$ be continuous. For every compact $K \subset H^s(X)$ and every $\varepsilon > 0$, there exist N_X, N_Y and a ReLU network $G_{\mathrm{NN}} : \mathbb{R}^{N_X} \to \mathbb{R}^{N_Y}$ such that*

$$\sup_{u \in K} \left\| \mathcal{G}(u) - R^Y_{N_Y} G_{\mathrm{NN}} P^X_{N_X} u \right\|_{L^2(Y)} \le \varepsilon. \tag{60}$$

(II) ***Quantitative fixed-metric bound.*** *Assume additionally that $\mathcal{G}$ satisfies the L^2-Lipschitz condition (9), that $\mathbb{E}_\mu \|u\|^2_{H^s(X)} \le B$, and that $E^Y_\mu = \mathbb{E}_\mu \|\mathcal{G}(u)\|^2_{H^t(Y)} < \infty$. Then*

$$\begin{aligned}\mathbb{E}_\mu \|\mathcal{G}(u) - R^Y_{N_Y} G_{\mathrm{NN}} P^X_{N_X} u\|^2_{L^2(Y)} \le 3\Big(& L^2_{\mathcal{G}} C^2_X B N_X^{-2s/n_X} + \varepsilon^2_{\mathrm{NN}} \\ & + C^2_Y E^Y_\mu N_Y^{-2t/n_Y} \Big).\end{aligned} \tag{61}$$

(III) ***Basis optimality.*** *For each compact closed n-manifold M,*

$$d_N(B_{H^s(M)}, L^2(M)) \asymp N^{-s/n}, \tag{62}$$

and the first N Laplace–Beltrami eigenfunctions achieve this rate.

(IV) ***Manifold aliasing.*** *If $s > n/2$ and $f, h \in H^s(M)$, then*

$$\|P_N(fh) - P_N(P_N f\, P_N h)\|_{L^2(M)} \le C_{\mathrm{al}} N^{-s/n} \|f\|_{H^s} \|h\|_{H^s}. \tag{63}$$

Proof. Part (I) is the compact-set existence theorem. Part (II) is the quantitative fixed-metric bound. Part (III) is the Sobolev-ball Kolmogorov-width theorem. Part (IV) is the bilinear manifold aliasing lemma. □

11 Conclusion

The Beltrami–Laplacian construction gives a manifold-native analogue of Fourier spectral neural operators. The fixed-metric theory requires neither periodic Euclidean coordinates nor toroidal topology. It rests instead on the spectral theorem for the Laplace–Beltrami operator, Weyl asymptotics, Sobolev truncation, finite-dimensional neural approximation, and classical width theory.

The strongest rigorous conclusion is therefore a fixed-metric approximation theorem: once a compact Riemannian geometry is specified, the BL eigenbasis gives a complete spectral encoder–decoder, quantitative Sobolev truncation rates, and optimal linear approximation of Sobolev balls. Learned metrics, Darcy-type PDE emulation, and atlas-based implementations are natural extensions, but they require additional assumptions and should not be folded into the main theorem without separate perturbation, regularity, and discretisation arguments.

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